

ing soft items automatically in their baggage.

“Besides, it is very easy to use by connecting with micro-USB and power sources like a plug or power bank. We use aerodynamic technology to design our product, and the size of the current product is only thirty per cent of the prototype, yet the vacuum power is 2.7 times of the household vacuum machine. This design has already earned us many patents from all over the world.”

With a diameter of 36mm and a height of 70mm, the device weighs about 77 grams. Vacuum bags used with it are manufactured by the company itself, and come with a slider and a double zipper. A high-strength polymer material is used for the bags to handle the pressure easily.

The bags are available in three sizes: small (36cm × 36cm),

medium (50cm × 60cm) and large (70cm × 100cm). Once the vacuum bags are filled with travel paraphernalia, they need to be zipped such that there are no air gaps.

The product uses a special kind of adaptor to connect with the bag. On pushing the power button, the vacuum motor starts and the process completes within a few minutes, depending on the amount of stuff inside the bag. The smart sensor present inside the device detects the pressure and stops on its own after the process is complete.

While working, it makes a low, 50dB noise. Its light bar turns from red to green once vacuuming is complete. The user can then unscrew the product. To take out the items, one can simply open the bag, when soft items will start to expand back to their original size and shape.

An advantage of this product is that vacuum prevents odour-emitting bacteria in worn clothes from growing and producing a foul smell. It is available in white, black, purple and pink colours.

Chao says, “We are currently focusing on B2B model and are looking for distribution channels all over the world to expand our market. VAGO has already sold in more than twenty countries and has collaborated with many well-known channels.

“In the future, we are planning to collaborate with more travel-related, well-known 3C stores (electronic stores) and airline-related channels. We are also planning to collaborate with travel-related influencers, and join exhibitions to expand our market and brand awareness in India.”

—Ayushee Sharma

4 SATELLITE DATA ENABLING BETTER DECISION-MAKING

Use of satellite data helps governments and industries make better decisions at the right time and provide new services. ICEYE has developed its synthetic-aperture radar (SAR) sensor technologies suitable for satellites weighing under hundred kilograms. It has launched a small satellite SAR imaging constellation, which is customised for high revisit rate and frequent delivery of high-resolution imagery.

The satellite can provide accurate data of any location on Earth, regardless of weather conditions, through cloud cover and darkness. This enables regular monitoring of vast sea, large land areas and atmosphere, and frequent imaging of hard-to-reach regions. ICEYE satellite constellation provides new images of the same location every hour, tracking all changes that happen.

X-band radar imaging is available at 1x1-metre resolution as Spotlight, at 1.5x1.5-metre and 3x3-metre resolutions as Stripmap, and 20x20-metre resolution as ScanSAR. The initial six-satellite constellations

for 2019 were launched in orbit in three different inclinations, with two satellites in each inclination. This provides different angle imaging multiple times a day for specified areas of interest.

ICEYE SAR processor compensates the following effects: range spread loss, elevation antenna pattern, sensor settings variation (receiver gain, transmit power, duty cycle), effects of different azimuth and range bandwidths.

Remote sensing devices with electro-optical (EO) sensors can be carried with various platform groups: space-borne includes satellites, shuttles or space stations; airborne includes manned or unmanned aircraft, aerial vehicles, etc; and ground-based includes installed sensors, mobile devices, animals or humans.

Fundamental differences between EO technologies depend on whether data is collected through a passive or an active remote sensing system. Passive remote sensing monitors solar energy, which Earth's surface

or atmosphere reflects and re-emits into space. This technology delivers data only in sunlight and cloud-less circumstances.

Commonly-used passive sensors include optical and thermal sensors, while active sensors include radar and laser technologies. Active remote sensing measures the energy received back from the Earth's surface or atmosphere.

There are two types of radar-based systems that are used for satellite-based microwave imaging: real aperture radar (RAR) and SAR. SAR can penetrate clouds and does not require sunlight to provide reliable remote sensing data.

Machine learning algorithms are used to scan huge data masses of satellite imagery and to develop models to extract features, detect changes and predict upcoming situations with the help of artificial intelligence (AI).

Volume of applications is huge, and it is rapidly increasing. Given below are some examples of satellite data uses: